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Conference Paper · January 2024

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An IoT Based Smart Hydroponics – Plant Automation BOT

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Abstract — Traditional Agriculture is unpredictable and very desirable. The sector has many problems like limited resources, financing, labour and land. With these problems, it becomes harder to get high yields. This is the present case in India, where the condition of farming is much better in developed western countries but even they are subjected to similar problems. Over the years, there is a need for increased output from agriculture to meet the needs of the population. Luckily, there have been alternative ways to traditional agriculture. One such different method is Hydroponics. Over the years, Hydroponics is a type of method of growing plants on less land and with fewer resources. Present hydroponics farms require some human interactions. We propose an automation solution for these Hydroponics systems. This method of farming reuses the majority of water and so saves around 90% of the water. Adding automation into the picture can help scale farms without worrying about the workforce and additionally data received.

Keywords — *Hydroponics, IoT, Life-long learnings*

I. INTRODUCTION

The science of hydroponics began with experiments to discover the basic nutrient requirements of plants and the ways in which plants process those elements. That knowledge has evolved into the preference to automate the delivery of the nutrient mix for the plant to optimize development and adjust the nutrient for specific plants, and specific stages of plant growth. For instance, we understand plants use more Nitrogen throughout the green growth phase when the plant is young, and an increased amount of Phosphorus throughout the fruiting and flowering stage. Additionally, plants will tend to do whatever they need to survive when they are in less than desirable conditions. These less desirable conditions reduce the growth and tend to seriously affect the product quality and flavour. Plant managing waste has become a very difficult task in today's world. As the population on earth is increasing day by day, the amount of garbage is simultaneously increasing at a rapid rate. If appropriate measures are not taken immediately for the collection,

segregation, and disposal of garbage, the earth will be covered with more garbage and less soil. [6] Nutrients for the shops are supplied to the roots in the form of results that can be either in the form of static or flowing. Hydroponics can be cultivated both in greenhouse and glasshouse terrain.[7] The hydroponics farming system is a new technology that allows farmers to develop their land outside the need for soil and get better results.



Fig. 1: Manual Hydroponics Farming

II. OBJECTIVE OF THE PROJECT

Evolve protocols for imposing named cultivars of leaf, fruit, and seed-yielding specialty vegetables, sauces, and spices, and to decide agents affecting their adjustment to maintainable hydroponic civilization under controlled area husbandry (CEA) systems. To evaluate the result of dissimilar aggregate media on growth, yield, and merchantable quality of named fruits, vegetables, herbs and spices.

III. LITERATURE REVIEW

Vertical Farming is an agricultural approach regarding large-scale meal manufacturing in high-upward thrust homes that allows the rapid boom and deliberate manufacturing controlling environmental situations and nutrient answers to plants primarily based totally on hydroponics the use of modern greenhouse techniques and technology. According to Perez's studies, vertical farming consists of each discipline of engineering and herbal sciences. Recently, generating sustainable meals in towns

has garnered a lot of hobbies and interest in lots of educational and realistic fields. While there are nevertheless restricted daunting technical and realistic program issues concerning imposing vertical farming. However, at present, those farms are in large part developing and conveying specific kinds of plants in interior towns consisting of China, Holland, South Korea, Japan, Canada, Italy, the U.S, Singapore, United Arab Emirates, and England [1]. As clever machines and sensors crop up on farms and farm statistics develop in amount and scope, farming procedures turn into an increasing number of statistics-pushed and statistics-enabled. Rapid trends inside the Internet of Things (IoT) are propelling the phenomenon of what is referred to as Smart Farming. Monitoring and controlling agricultural manufacturing and feed through the use of superior sensor structures are similar to programs of IoT. Such structures will make certain the fitness of plant beginning merchandise supposed each for human and animal consumption. Although IoT is gaining momentum to allow generation for developing a ubiquitous computing surrounding, unique issues are required to manner massive quantities of statistics originating from and circulating in, this sort of allotted and heterogeneous surroundings. On the opposite hand, hydroponics is one of the farming technology that became taken into consideration because the fastest developing quarter of agriculture and may govern meal manufacturing in the future. Hydroponics farming completely offers the proper quantity and sort of vitamins that the flowers want at a proper time and maybe established interior to maximize be had space. Soil-associated issues had been solved through hydroponics like developing flowers that wished hard-to-keep soil situations, minimum weeding, and less difficult to harvest. Combining hydroponics with IoT, could cause a successful crop harvest and doubtlessly enhance boom quality [2]. The emergence of IoT has allowed farmers to automate hydroponic culture. Monitoring of water stage, pH, temperature, flow, and mild depth may be done, and they may be regulated through the use of IoT. So a pretty quantity of studies has been executed through using IoT for tracking and controlling the hydroponic machine. In hydroponics, so accordingly, microcontrollers have a Real-Time clock that might manipulate lights through 18 methods of the relay switch. This Pi3 acts as an interface between the microcontroller and the cloud. The Pi3 controls the hydroponic surroundings primarily based totally on the statistics captured from sensors. [3]. The importance of hydroponics is in supplying a manner for a median character to develop their meals without the want of soil, Correct pH stage, air temperature, relative humidity, PH stage, nutrient stage of the water, and accurate irrigation of water is severely critical in hydroponics. Therefore, the assistance of a control machine

that video displays units of those elements is precious and could make certain better fulfilment and performance fees for the grower. Hydroponics Management System (HMS) is a hydroponics machine that allows customers to manipulate positive mechanisms for refilling, sprinkling, draining, and plenty of greater via the internet utility. They also can display the pH stage, relative humidity, air temperature, and water stage, that is statistics accrued from the sensors. As long as there's a web connection to be had, the person must be capable of displaying and manipulating those each time and anywhere. Through HMS, hydroponics must turn out to be greater available and to be had. This will permit busy human beings to develop their meals whilst coping with and tracking the machine [4]. Hydroponics is a method of developing flowers in nutrient answers without or with the usage of an inert medium consisting of gravel, vermiculite, rock wool, peat moss, noticed dust, coir dust, coconut fibre, etc. to offer mechanical support. The period Hydroponics became derived from the Greek phrases hydro' method water and pones' method labour and method water work. The phrase hydroponics became coined by Professor William Gericke in the early 1930s; describes the development of flowers with their roots suspended in water containing mineral vitamins [8]. Researchers at Purdue University advanced the nutria culture machine in 1940. Due to fast urbanization and industrialization, now no longer best is the cultivable land reducing however additionally traditional agricultural practices inflict an extensive variety of bad influences on the surroundings. To sustainably feed the world's developing population; techniques for developing enough meals should evolve. Modification inside the booming medium is an opportunity for sustainable manufacturing and to preserve rapidly depleting land and to be had water resources. In the existing scenario, soil-much less cultivation is probably started out efficiently and taken into consideration as an opportunity choice for developing wholesome meals flowers, plants, or vegetables (Butler and Oebker, 2006). Agriculture without soil consists of hydro agriculture (Hydroponics), aqua agriculture (Aquaponics), and cardio agriculture (Aeroponics) in addition to substrate culture [5].

IV. METHODOLOGY

1. The esp32 here is the brain of the whole system. It controls all the communication sensors and the telegram app.
2. A Chatbot is created to send commands and receive their responses. Telegram's Bot API was used for making the chatbot since telegram provided a friendly user interface and the bot can be made with no charges.

3. The ESP32 will check if any new message comes with the chatbot, this check is done frequently every two seconds.
4. On receiving a message The ESP looks within defined messages, if the command here is not defined an “unidentified command” is replied. If the command is defined it will perform the respective action. There are three commands defined for this project.
5. The Status command reads Moisture data from the moisture sensor, Humidity and Temperature data from the DHT11 sensor module.
6. R1 Command turns on the feedback pump in the residual water tank to send water from it back to the main water tank. The reuse of this water ensures reduced wastage of water and circulates 90% of the water throughout the system.
7. R2 command turns off the valve to let water flow to the plants. The command overrides the present status of the valve to the user-specified. The valve is also automatically turned on/off in response to the moisture data.

The overall detailed technologies and protocols used in the development of this project are described below

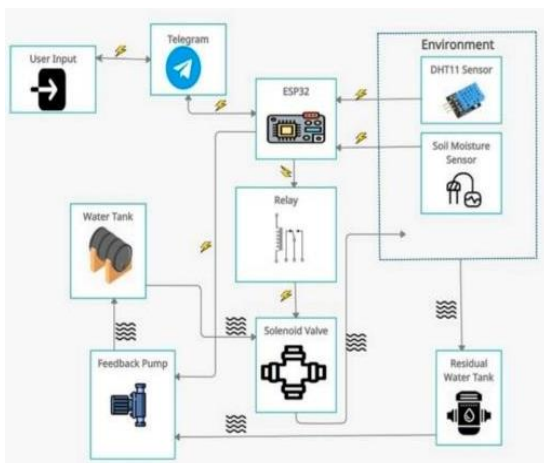


Fig 2.: Block Diagrams of hydroponics bot

V. RESULTS & DISCUSSIONS



Fig 3.: Hydroponic plant auto Bot and Arial view

The Developed product can operate autonomously, without human intervention. And monitoring of data is successfully done. This data can be viewed in the telegram app with its corresponding chatbot. Additionally, certain aspects of the system are also in control via the chatbot like turning on/off the relays and pumps. To control and get the real-time data commands are defined by the user, and commands such as “Status”, “R1” and “R2” as shown in Fig.12 are defined for this project for data, controlling of the pump and controlling of valve respectively. More commands can be added to the list based on the number of plants, sensor and actuators.

The inclusion of the residual water tank has proved to be very useful in reusing the water that if not would have been wasted. As a result refilling of the tank becomes less frequent. The product can be very useful for both hobbyists and agriculture industries alike.



Fig.4: Demonstration of Hydroponics Bot

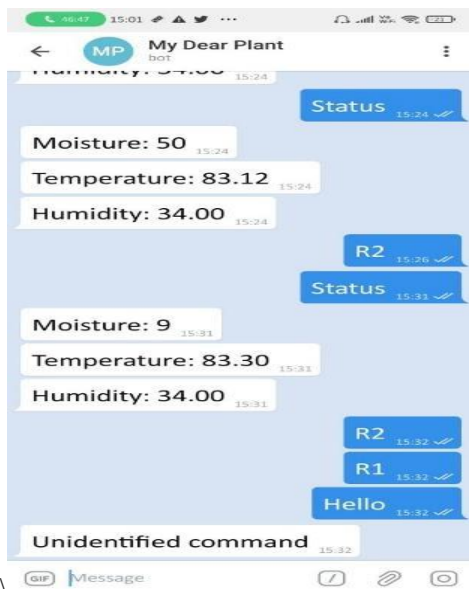


Fig.5 Telegram output



Fig.6 Hardware Setup

VI. CONCLUSION

Hydroponic culture is the most intensive technique of crop production in today's agricultural industry. It is highly productive, conservative of water, land and space, and protective of the environment. The automation solution increases the yield of an already efficient system. This solution makes it easier to scale the farms without worrying about the direct need for manpower. Instead, the manpower can be spread across other manual jobs. The application of IoT and Bigdata significantly improves the system. an experimental perspective has grown as a successful community-based technology support planting system. The need is greatly satisfied by the production in the system. The commercial success of the system is due to its flexible nature to adapt to an ample number of environmental changes. Say, suppose the implementation of hydroponics in a semi-arid land will surely have variations in comparison to implementation in a tropical land. Thus, a sound production system can only be helpful for cultivation by adapting itself to upgrading technology.

ACKNOWLEDGMENT

We like to thank our Institute for supporting us in the development of this project and the faculty who also guided us special thanks to our faculty advisor Mr. Santosh Naik who has helped, given suggestions and encouragement, and has relentless guidance and assisted us to coordinate our paper.

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